

1. Introduction

Flooding is one of the most frequent and devastating natural disasters in Bangladesh.

This project proposes a Smart Flood Prediction & Evacuation System that uses real-time data, predictive analytics, and intelligent routing algorithms to minimize loss of life and property.

2. Objectives

- Predict flood risks using historical and real-time weather data
- Identify and visualize high-risk zones dynamically
- Provide optimized evacuation routes using algorithms such as A*
- Simulate flood scenarios using a What-if engine
- Send alerts via SMS/email (simulated)

3. Problem Statement

Existing systems lack real-time prediction, dynamic routing, and personalized alerts.

This system aims to address these gaps using modern technologies and intelligent system design.

4. System Overview

Core modules include:

- Flood Prediction Engine
- Risk Zone Mapping System
- Evacuation Route Optimizer
- Alert System
- What-if Simulation Engine

5. Technology Stack

Backend: Spring Boot (Java)

Frontend: next.js

Database: MySQL/PostgreSQL

APIs: OpenWeatherMap

6. Implementation Plan

Phase 1: Backend setup

Phase 2: API integration

Phase 3: Algorithm implementation

Phase 4: Simulation and alerts

7. Additional Features (Advanced)

- Real-time dashboard for monitoring flood conditions
- Bangladesh-specific river and district-level data integration
- Smart alert prioritization based on user location and risk level
- Multi-route evacuation system (safest, fastest, least congested)
- Shelter and resource recommendation system
- Offline functionality for low connectivity scenarios
- Basic AI-based flood prediction using historical data
- Interactive simulation mode for scenario analysis
- Community-based reporting system for real-time updates

8. Expected Outcome

A real-time intelligent system for flood prediction, risk mapping, and evacuation planning.

9. Impact

This system supports disaster management efforts in Bangladesh and helps reduce loss of life and property.

It provides a scalable and practical solution tailored to local challenges.

10. Future Enhancements

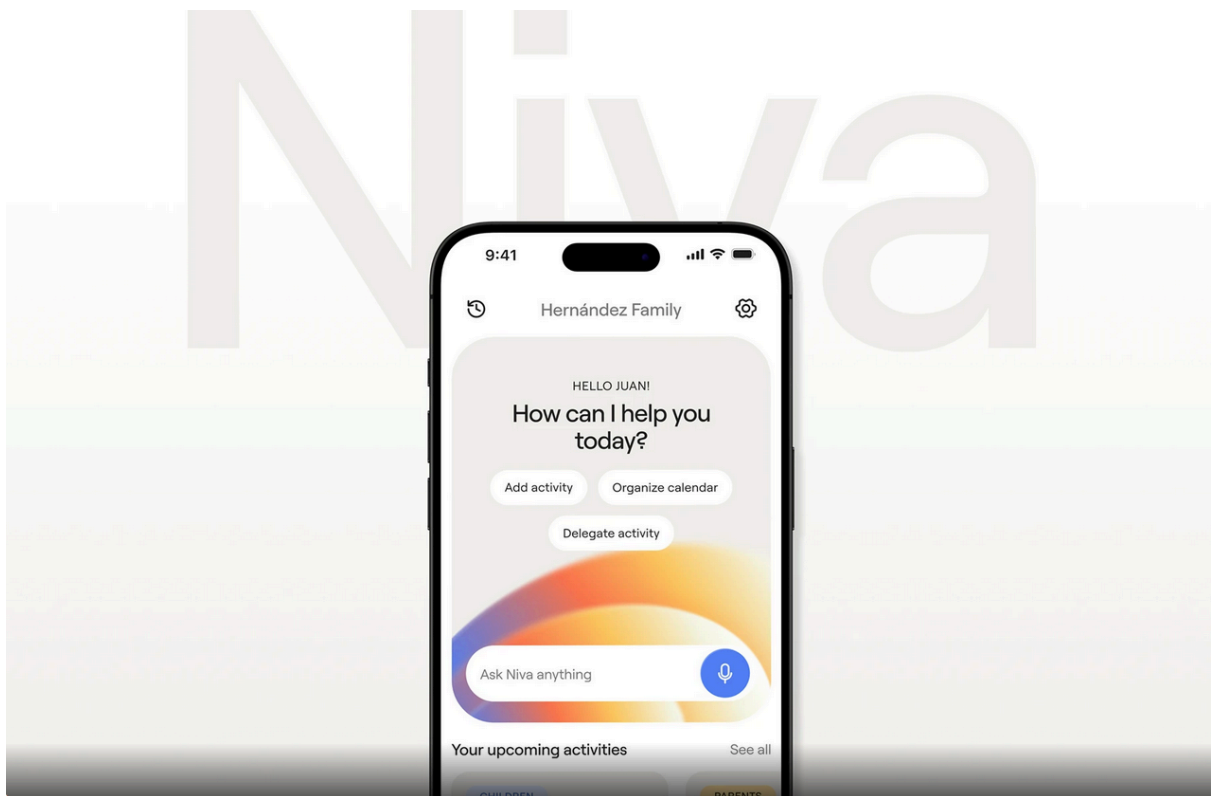
- Advanced AI/ML integration
- Mobile application development
- Integration with real SMS and emergency services
- Drone-based flood monitoring

11. Conclusion

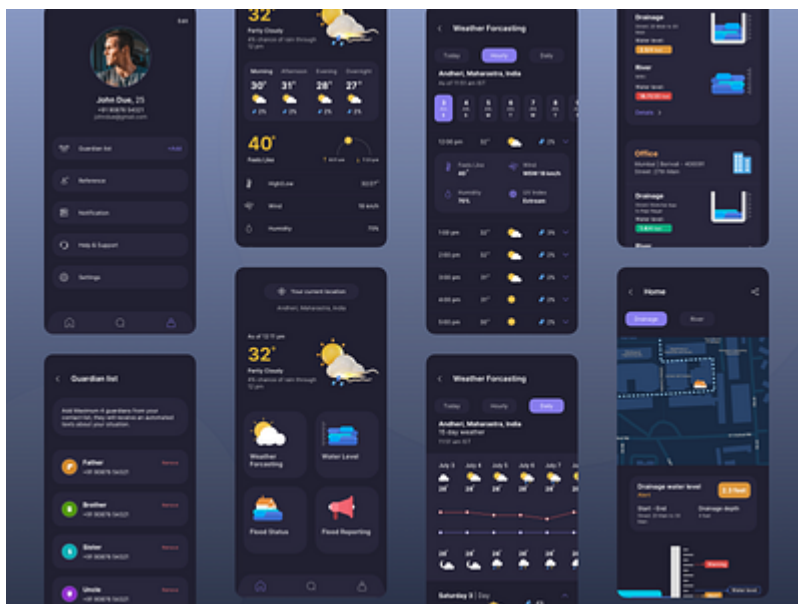
The proposed system combines modern software engineering practices with real-world problem-solving.

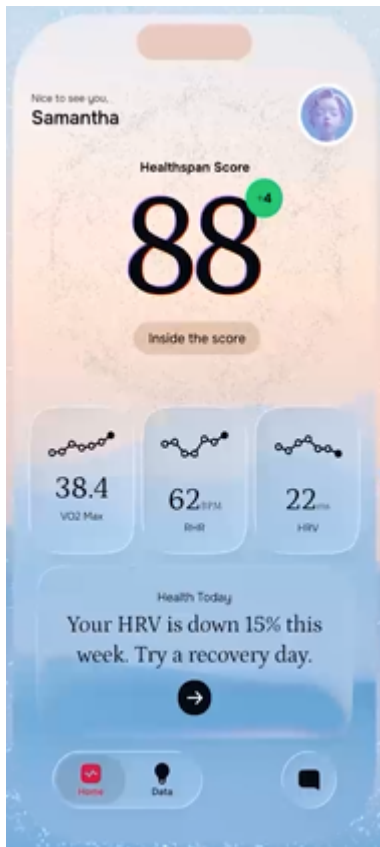
It demonstrates a scalable and impactful approach to disaster management using object-oriented design principles.

UI:

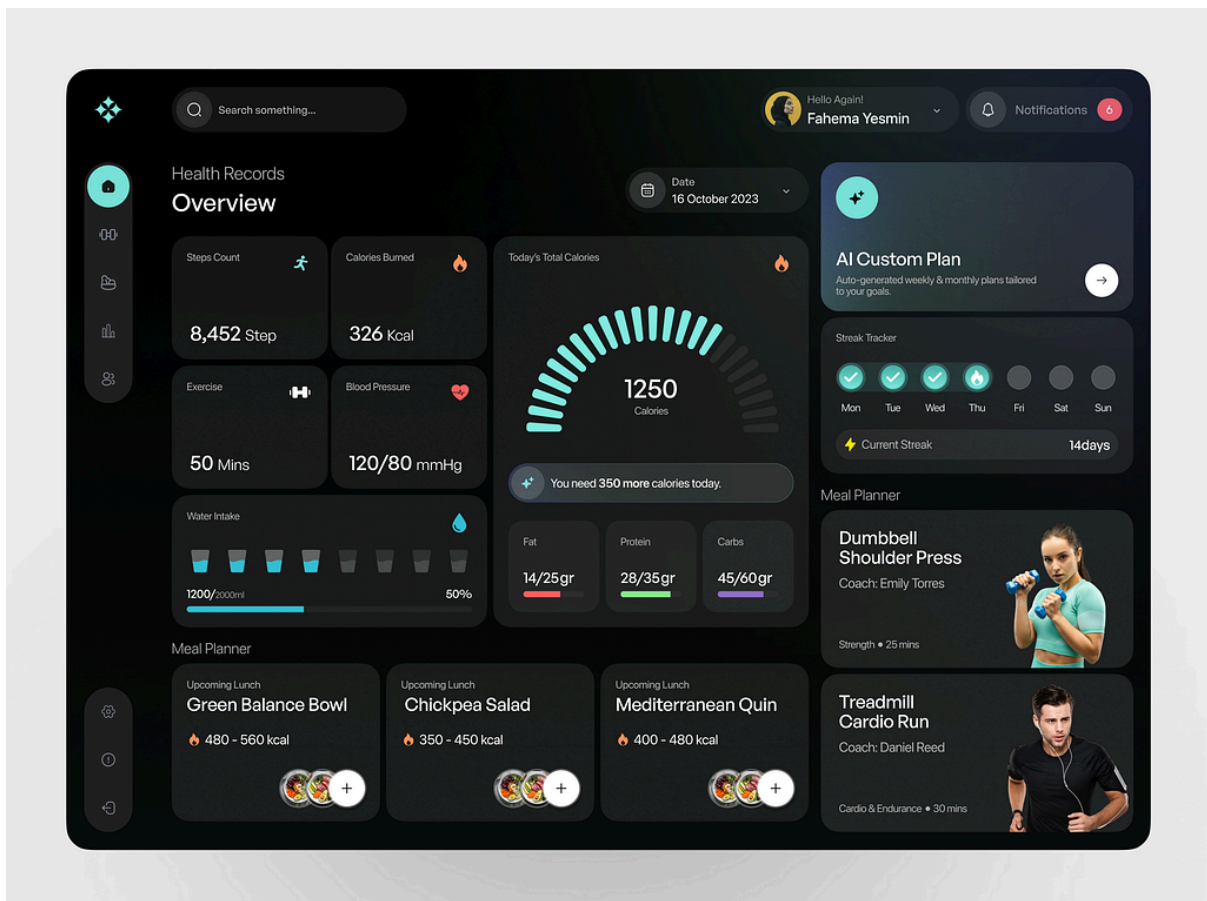


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